

Ex. If $f(x) \in F[x]$; F is field is a polynomial of degree ≥ 1
 then the quotient $\frac{F[x]}{\langle f(x) \rangle}$ is a field as $\langle f(x) \rangle \triangleleft F[x]$ is maximal

If R is a commutative ring with 1

$I \triangleleft R$ (ideal) i.e. $I \subseteq R$ & $(I, +) \leq R$

& $r \in R, a \in I \Rightarrow ar \in I$

$$\frac{R}{I} = \{ I+r \mid r \in R \}$$

$$(I+r) + (I+s) = I+(r+s)$$

$$(I+r)(I+s) = I+rs$$

$I+1$ is the unity in the ring $\frac{R}{I}$ called the quotient ring of R by I

$$I+a = I \text{ iff } a \in I$$

$$I+a = I+b \Leftrightarrow a-b \in I$$

Ideals of $\frac{R}{I}$: If $J \triangleleft R$ s.t. $I \subseteq J$; then

$$\frac{J}{I} \triangleleft \frac{R}{I}$$

Note that I is the zero of $\frac{R}{I}$

$$I + (I+a)^2 = I+a$$

$$\eta: \frac{\theta \triangleleft R}{I} ; \text{ then } \exists \frac{J \triangleleft R}{I} \text{ s.t. } \theta = \frac{J}{I}$$

Quotient map $R \xrightarrow{\eta} \frac{R}{I}$ (onto map)

$$\pi \mapsto I + \pi$$

η is a ~~linear~~ homomorphism

$$\eta(a+b) = \eta(a) + \eta(b) \quad \eta(ab) = \eta(a)\eta(b)$$

$$\ker(\eta) = I$$

Thm $\eta: R \xrightarrow{\text{surjective}} \frac{R}{I}$ is a hom. then for any ideal $\theta \triangleleft \frac{R}{I}$, $\phi^{-1}(\theta) \triangleleft R$

$$x, y \in \phi^{-1}(\theta) ; \phi(x), \phi(y) \in \theta$$

$$\theta \triangleleft \frac{R}{I} ; \phi(x+y) = \phi(x) + \phi(y) \in \theta$$

$$\Rightarrow x+y \in \phi^{-1}(\theta) \quad (1) \triangleleft R$$

$$x \in R, y \in \phi^{-1}(\theta) \quad \phi(xy) = \phi(x)\phi(y) \in \theta \text{ as } \theta \text{ is an ideal } \triangleleft \frac{R}{I}$$

$$\Rightarrow xy \in \phi^{-1}(\theta) \quad (2) \triangleleft R$$

$$(1) \& (2) \Rightarrow \phi^{-1}(\theta) \triangleleft R$$

$$R \xrightarrow{\eta} \frac{R}{I} \triangleleft \frac{\theta \triangleleft R}{I}$$

$$\eta^{-1}(\theta) \triangleleft R, \text{ call it } J$$

$$\text{then } \frac{J}{I} = \eta(J) = \theta$$

$$\mathbb{F} \rightarrow \frac{\mathbb{F}[x]}{\langle f(x) \rangle}, \quad f(x) \text{ is irreducible}$$

$a \mapsto I + a$ when $I = \langle f(x) \rangle$
 is injective & realize $\mathbb{F}$ inside $\frac{\mathbb{F}[x]}{\langle f(x) \rangle}$ as $\eta(\mathbb{F})$

$$K = \frac{\mathbb{F}[x]}{\langle f(x) \rangle} \text{ is an extension of } \mathbb{F}$$

(K is an extⁿ of $\mathbb{F}$)

Recall: if K is an extⁿ of $\mathbb{F}$ then K is a vector space over $\mathbb{F}$

$$\text{if } K = \frac{\mathbb{F}[x]}{\langle f(x) \rangle}, \text{ then } \dim_{\mathbb{F}} K = \deg(f(x)) = n$$

$$\text{in fact: } \{I + 1, I + x, I + x^2, \dots, I + x^{n-1}\}$$

$$I = \sum a_i(I + x^i) = I + \sum a_i x^i$$

$$\sum a_i x^i \in I = \langle f(x) \rangle$$

$$\deg(\sum a_i x^i) < n$$

$$\Rightarrow a_i = 0 \quad \forall i \text{ as otherwise } \deg > n$$

$$\text{let } I + g(x) \in \frac{\mathbb{F}[x]}{\langle f(x) \rangle}$$

$$\langle f(x) \rangle$$

By division

$$g(x) = f(x)q(x) + r(x)$$

$$r = 0 \text{ or } \deg(r(x)) < n$$

$$I + g(x) = (I + f(x)) (I + q(x)) + (I + r(x))$$

$$= I + r(x)$$

$$\text{if } r(x) = \sum_{i=0}^{n-1} a_i x^i$$

$$I + g(x) = \sum_{i=0}^{n-1} (I + a_i) (I + x^i) = \sum_{i=0}^{n-1} a_i (I + x^i)$$